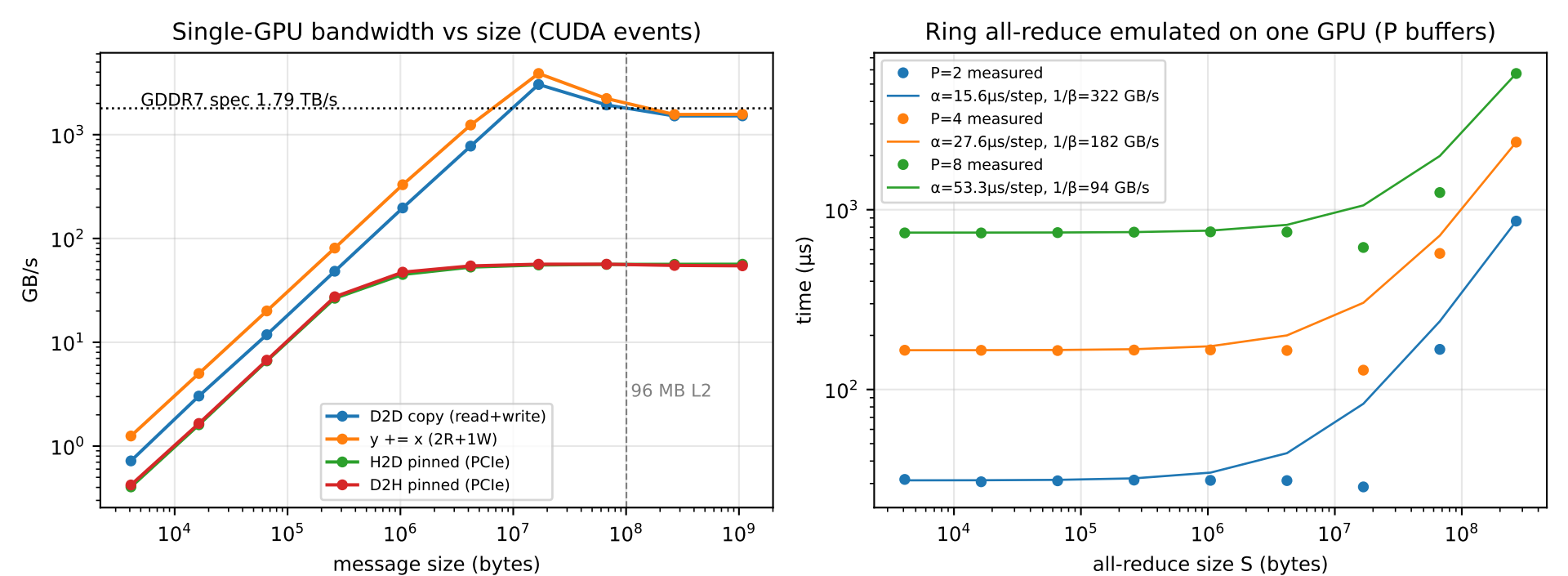
Single-GPU bandwidth vs size (CUDA events)
GDDR7 spec 1.79 TB/s
96 MB L2
D2D copy (read+write)
y += x (2R+1W)
H2D pinned (PCIe)
D2H pinned (PCIe)
GB/s
message size (bytes)
Ring all-reduce emulated on one GPU (P buffers)
P=2 measured
α=15.6µs/step, 1/β=322 GB/s
P=4 measured
α=27.6µs/step, 1/β=182 GB/s
P=8 measured
α=53.3µs/step, 1/β=94 GB/s
time (µs)
all-reduce size S (bytes)